UNIFIED ADS MANAGEMENT PLATFORM

Shared UI vs provider-specific workflows and services

Purpose: define, feature by feature, what should be built once and what must have separate Google Ads and Meta Ads logic. This is a planning blueprint; it does not modify the existing codebase.

LABEL	MEANING	EXAMPLE
SHARED UI + SHARED SERVICE	Build once and reuse across both platforms.	Workspace, team roles, connection registry, dashboard shell, audit log.
SHARED UI + SEPARATE ADAPTERS	One screen, but backend routes to the correct provider integration.	Campaign list, reporting, pause/resume, status display.
SHARED SHELL + SEPARATE FORMS	Same navigation and general layout, but different provider-specific inputs.	Campaign creation, bidding, targeting, ad group/ad set editors.
SEPARATE PROVIDER WORKFLOW	Different setup steps, data models or permissions; don't force one identical editor.	Google OAuth vs Meta OAuth; Google conversion actions vs Meta Pixel/Dataset setup.

1. The most important design decision

Do not build two complete websites. Build one SaaS product with a shared user experience and two separate provider adapters. Use separate provider-specific screens only when the fields, workflow, permissions, or underlying concepts really differ.

Visual companion: See the separate architecture diagram for the end-to-end flow and the dedicated connection diagram for the Google-versus-Meta OAuth steps.

2. Connect Ad Accounts: exact UI and service split

Recommendation: one Connections page in the product, with two independent “Connect” actions. The user should not need to understand the backend split, but the implementation should preserve separate provider OAuth handlers, permissions, account discovery, token lifecycle, and API clients.

STEP	SHARED / COMMON	GOOGLE ADS PATH	META ADS PATH
1. Click Connect	One Connections page, one consistent button and success/error UI.	Connect Google Ads starts Google OAuth 2.0 authorization.	Connect Meta Ads starts Meta’s OAuth/app authorization flow.
2. Consent and permissions	Shared explanation, loading state and cancel behavior.	Request the required Google Ads API scope and user consent. OAuth app credentials belong to your Google Cloud project. [G1]	Request only the required Meta ads permissions. Access/approval level depends on whether the app serves only its own account or external customers. [M1]
3. OAuth callback	Shared callback-result handler pattern, telemetry and user-facing error model; do not use a single provider callback URL handler that guesses the provider.	Separate route/service exchanges the code, requests offline access when appropriate, and handles refresh-token lifecycle. [G1][G2]	Separate callback/service handles the token response and its validity/expiration rules. Keep Meta app credentials separate from Google credentials. [M1][M2]
4. Store credentials	Shared encrypted credential vault and tenant scoping. Never expose tokens to frontend JavaScript.	Store Google authorization securely; track refresh status and account context. Google API calls also require the Google Ads developer token. [G2]	Store Meta authorization securely with permission and expiry/validity metadata; keep it scoped to the customer/workspace. [M1][M2]
5. Discover accounts	Same account-selection component: list, search, select, show name and status.	Google adapter fetches accessible customer IDs and any needed manager-account context. [G1][G2]	Meta adapter fetches accessible businesses/ad accounts and displays the selected ad account ID (often represented as act_...). [M1]
6. Validate connection	Shared “Test connection” service contract and status values: Connected / Needs attention / Reconnect required.	A safe read request confirms permissions and customer access; handle developer-token, customer-ID and quota errors.	A safe read request confirms ad-account access and granted permissions; handle permission, token and rate-limit errors.
7. Persist & show connected	One connection registry row/card, account selector and reconnect/disconnect action.	Store provider=google_ads, external customer ID, token reference, permissions and sync health.	Store provider=meta_ads, external ad-account ID, token reference, permissions and sync health.

What must be separate for account connection?

COMPONENT	SHARED OR SEPARATE?	REASON
Connections page / buttons / status cards	SHARED	Same customer-facing interaction and common status vocabulary.
OAuth start + callback handlers	SEPARATE	Different identity providers, scopes, callback handling and token formats/lifecycle.
Credential vault & encryption framework	SHARED INFRASTRUCTURE	Same secure storage system, but separate credentials, provider type, and access metadata per connection.
Account discovery functions	SEPARATE ADAPTER METHODS	Different account hierarchy and identifiers.
Connection record schema	SHARED BASE + PROVIDER FIELDS	Common workspace/provider/status fields plus platform-specific account IDs/permission metadata.
Connection diagnostics page	SHARED PAGE, PROVIDER-SPECIFIC DIAGNOSTICS	One UX; the error mapping/help links need to reflect the provider.

3. Every product area: reuse decision matrix

This is the build decision for each major feature. “Shared” means the user-facing product experience is reusable; it does not imply that the Google and Meta API calls or data schemas are the same.

FEATURE	UI DESIGN	SHARED SERVICE / LOGIC	SEPARATE PROVIDER WORK / RULES
Login, workspace, team roles	Shared UI + shared service	Identity, tenant isolation, roles, invitations, workspace settings.	Provider connection permissions are checked separately.
Connections & account picker	Shared UI + shared orchestration	Connection cards, selection state, connection registry, status and reconnect UX.	Google OAuth/account discovery/token refresh vs Meta OAuth/permissions/business-account discovery.
Dashboard and KPIs	Shared dashboard; separate adapters	Date filters, chart components, shared metrics model, caching and freshness labels.	Provider reports are fetched separately; currency, timezone, attribution and definition may differ.
Campaign list and actions	Shared table + separate adapters	Search, filters, bulk-action UX, confirmation, audit trail, normalized status vocabulary.	Separate list/mutate API calls; test support for each campaign type and operation.
Create/edit campaign	Shared wizard shell + provider-specific steps	Common name, draft state, validation framework, progress indicator.	Google campaign types/network/bidding versus Meta objectives/special ad categories/placements/optimization. Use separate form modules.
Budgets and bids	Shared field patterns, separate validators/adapters	Currency formatting, input widgets, change review, audit history.	Budget periods, budget placement and bidding strategy choices vary; expose only supported combinations.
Ad group / ad set	Shared navigation; separate editors	List/search, breadcrumbs, save/cancel layout.	Google Ad Groups and Meta Ad Sets have different relationships, targeting and optimization fields; not one shared data model.
Ads and creative	Shared library; separate ad editors	File storage, asset tags, previews, upload queue, naming and deduplication.	Provider-specific creative structures, placements, formats, text limits, destination requirements and review states.
Audiences / targeting	Shared section and basic inventory	Common browse/search shell and neutral labels.	Separate audience objects and targeting forms; not every type has a cross-platform equivalent.
Conversion tracking	Shared overview + event mapping; separate setup flows	Event dictionary, setup checklist, website notes, diagnostics UI, status reporting.	Google conversion actions/tag/GTM/offline conversion workflows differ from Meta Pixel/Dataset/events/Conversions API.
Reports and CSV export	Shared report builder/export UX	Filters, saved report definitions, export generation and download history.	Separate query shapes and metric normalization; mark unsupported metrics and keep platform attribution context.
Billing: our SaaS	Shared service and UI	Your subscription plans, invoice history, payment state, team/seat entitlements and app-level usage limits.	This is your product billing, independent of ad networks. Payment processor choice is a separate product decision.
Billing: ad accounts	Shared navigation only; separate provider details	Common summary cards only where definitions/permissions support them.	Google Ads billing API workflows have eligibility constraints such as monthly invoicing. Meta billing data/actions depend on its API and permissions. Do not promise a unified payment-method editor. [G3]
Conversions and event diagnostics	Shared diagnostics shell; separate validators	Common checks, severity levels, timestamp, help links and incident history.	Each adapter checks provider-specific event/tag state, permissions, and error codes.
Asset / media upload	Shared storage; separate upload services	Storage, hashes, metadata, content scanning, workspace access, queue UI.	Separate upload endpoints, IDs, accepted specs, processing status and constraints.
Notifications & audit logs	Shared service	Audit who changed what, operation status, retries, notifications and history.	Store provider request IDs/error detail and translate into safe, useful user messages.
API health and settings	Shared page; provider-specific checks	One health screen with last successful sync, permission status and reauth actions.	Google-specific developer-token/request/quota details; Meta token/permission/rate-limit/app-access diagnostics.

4. Conversion tracking: a shared overview, different setup journeys

CUSTOMER TASK	SHARED UX	GOOGLE ADS IMPLEMENTATION	META IMPLEMENTATION
See tracking status	One Conversion Tracking screen with provider tabs, status, last event and next step.	List available conversion actions and available status information.	List accessible Pixel/Dataset references and available event/diagnostic information.
Create/select tracking object	Shared setup checklist, owner, website, goal and status fields.	Google Ads conversion action/goal workflow.	Pixel/Dataset and event configuration workflow.
Install website tracking	Shared guide that asks for CMS/tag manager and who owns website changes.	Google tag or Google Tag Manager instructions; additional Google conversion options where supported.	Meta Pixel install instructions; optionally Meta Conversions API as a separate server-side implementation.
Map business event	Shared event dictionary: lead, purchase, signup, etc.	Map to the specific Google conversion action and its values/goals.	Map to Meta standard/custom events and any applicable custom conversion.
Verify events	Shared test checklist and diagnostics display.	Use available Google conversion diagnostics/reporting; note that reporting may be delayed.	Use available Meta test-event/diagnostics tooling; do not mark the website as installed merely because the Pixel object exists.
Show in dashboard	Shared conversion column labels only after mapping semantics.	Store source, conversion action, attribution context and currency.	Store source, event, attribution context and currency. Do not imply identical attribution.

Implementation boundary: connecting an ad account or creating a Pixel/conversion action does not automatically install tracking on the customer website. Installation requires website/CMS/tag-manager work or an authorized server-side integration. Keep this distinction visible in product status.

5. Billing: keep the two meanings separate

BILLING AREA	WHAT USER MEANS	WHAT TO BUILD
Your SaaS billing	What the customer pays your platform for using your software.	A normal subscription/billing module: plans, trial, invoices, payment state, seats and subscription changes. One shared service for both ad providers.
Google Ads account billing	What an advertiser owes Google / account budgets / invoices.	Separate Google-specific billing integration and eligibility checks. Official API billing workflows for billing setups, account budgets and invoices require monthly-invoicing setup. Otherwise provide read-only summaries or link to Google Ads UI where appropriate. [G3]
Meta Ads account billing	What an advertiser owes Meta / billing status / spend limits.	Separate Meta-specific data and actions where available and permissioned. For unsupported or permission-blocked actions, deep-link to Meta Ads Manager rather than inventing controls.
Unified spend reporting	How much ad spend is being reported across networks.	Shared summary UI, but label network, currency, account timezone, reporting date range and metric definitions. This is analytics, not the same thing as a unified invoice or payment method.

6. Suggested service boundary

SHARED SAAS CORE	GOOGLE ADS ADAPTER	META MARKETING ADAPTER
Users, workspaces, RBAC; connection registry; encrypted token vault abstraction; sync queue; retry scheduler; normalized reporting model; CSV exports; audit log; notifications; SaaS subscriptions.	Google OAuth flows; Ads client configuration; access-token refresh; accessible-customer discovery; campaign/ad group/ad methods; Google report queries; conversion-action/tag helpers; supported Google billing calls; Google error/quotas.	Meta OAuth and permission handling; business/ad-account discovery; token validity checks; campaign/ad set/ad/creative methods; Insights queries; Pixel/Dataset/event support; supported billing details; Meta error/rate-limit handling.

Security must-have: store tokens encrypted at rest, scope every credential to a workspace and provider, keep app secrets server-side, redact secrets from logs, support revoke/disconnect, and audit campaign-changing actions. Google's official credential guidance explicitly recommends secure storage and avoiding plaintext/transmitted or repository-stored secrets. [G2]

7. Recommended MVP build order

PHASE	BUILD	WHY
1. Safe foundations	Workspace, user roles, OAuth connections for both providers, token vault, account picker, connection diagnostics.	Prove both providers can connect reliably before adding more screens.
2. Read-only unified data	Dashboard, campaign list, basic performance metrics, date filters, account timezone/currency, CSV export.	Validate reporting and metric normalization without risking customer campaigns.
3. Safe campaign controls	Pause/resume, budget update where supported, confirmation dialogs, audit trail, partial-failure UI.	Adds useful management while keeping actions explicit and traceable.
4. Provider-specific create/edit	Separate campaign wizard steps, ad group/ad set editors, creatives and targeting.	Different schemas mean build provider modules, not a false one-size-fits-all form.
5. Conversion tracking	Google conversion setup; Meta Pixel/Dataset and optional CAPI; event mapping and diagnostics.	Tracking needs additional install/verification UX and permission/data-consent safeguards.
6. Billing and advanced operations	SaaS subscription billing; provider billing data/actions where supported; schedules, bulk operations and deeper reports.	Billing has different meanings and platform capability restrictions; add after core platform reliability.

8. Official references checked

[G1] Google Ads OAuth overview | [G2] Credential management
[G3] Google Ads billing overview | [G4] Google Ads campaigns overview
[M1] Meta Marketing API official Postman workspace | [M2] Meta Pixel implementation training